



Land & Water

	Overview	Land	Water	Databases & Software	News	Events	Resources	SOLAW 2021
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AQUASTAT

AquaCrop

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Crop Information

Tobacco

Maize

Sorghum

Wheat

Cotton

Alfalfa

Banana

Citrus

Grape

Pineapple

Potato

Groundnut

Olive

Safflower

Soybean

Sunflower

Bean

Pea

Sugarbeet

Sugarcane

Cabbage

Onion

Pepper

Tomato

Watermelon

Potato

This section presents information on water relations and water management of potato and provides links to other sources of information.

Crop Description and Climate

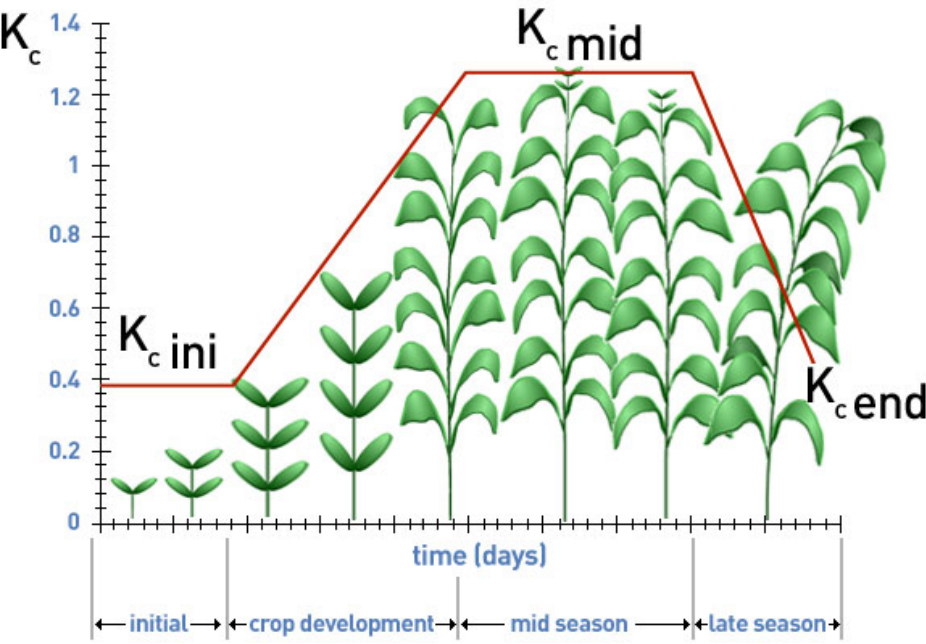
Potato (*Solanum tuberosum*) originates in the Andes from the tropical areas of high altitude. The crop is grown throughout the world but is of particular importance in the temperate climates. Present world production is some 308 million tons fresh tubers from 19 million ha. (FAOSTAT, 2001).

Yields are affected by temperature and optimum mean daily temperatures are 18 to 20°C. In general a night temperature of below 15°C is required for tuber initiation. Optimum soil temperature for normal tuber growth is 15 to 18°C. Tuber growth is sharply inhibited when below 10°C and above 30°C. Potato varieties can be grouped into early (90 to 120 days), medium (120 to 150 days) and late varieties (150 to 180 days). Cool conditions at planting lead to slow emergence which may extend the growing period. Early varieties bred for temperate climates require a daylength of 15 to 17 hours, while the late varieties produce good yields under both long or short day conditions. For tropical climates, varieties which tolerate short days are required for local adaptation.

Potato is grown in a 3 or more year rotation with other crops such as maize, beans and alfalfa, to maintain soil productivity, to check weeds and to reduce crop loss from insect damage and diseases, particularly soil-borne disease. Potato requires a well-drained, well-aerated, porous soil with pH of 5 to 6. Fertilizer requirements are relatively high and for an irrigated crop they are 80 to 120 kg/ha N, 50 to 80 kg/ha P and 125 to 160 kg/ha K. The crop is grown on ridges or on flat soil. For rainfed production in dry conditions, flat planting tends to give higher yields due to soil water conservation. Under irrigation the crop is mainly grown on ridges. The sowing depth is generally 5 to 10 cm, while plant spacing is 0.75 x 0.3 m under irrigation and 1 x 0.5 m under rainfed conditions. Cultivation during the growing period must avoid damage to roots and tubers, and in temperate climates ridges are earthed up to avoid greening of tubers.

The crop is moderately sensitive to soil salinity with yield decrease at different levels of ECe: 0% at 1.7, 10% at 2.5, 25% at 3.8, 50% at 5.9 and 100% at ECe 10 mmhos/cm.

The graph below depicts the crop stages of potato, and the table summarises the main crop coefficients used for water management.



CropWat	Crop characteristic Stage length, days	Initial	Stages of Development				Plant date	Region	
CLIMWAT			Crop Development	Mid-season	Late	Total			
Diagnostic Tools for Investment									
FAO Nile		25	30	30/45	30	115/130	Jan/ Nov May		
ETo Calculator		25	30	45	30	130			
GAEZ		30	35	50	30	145			
GeoNetwork		45	30	70	20	165	April Apr/ May Dec		
GLADIS		30	35	50	25	140			
HWSD									
Rapid Appraisal Procedure (RAP)		Depletion Coefficient, p:	0.5	>>	0.6	0.9	0.5		
SoiLEX	Root Depth, m	0.3	>>	>>	1.0	-			
SoiSTAT									
WaPOR	Crop Coefficient,Kc ¹ : with bare soil with grass cover	0.5	>>	1.15	0.5	-			
WaterLex									
	Yield Response Factor, Ky	0.2	0.8	-	1.0	0.85			

Water Requirements

Water Supply And Crop Yield

Water Uptake

Irrigation Scheduling

Irrigation Methods

Yield

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